1) Expand the following series to the fifth degree.

a) $\sum_{n=0}^{\infty} \frac{(-1)^n (x+2)^n}{2^n (2n)!}$

$1 - \frac{(x+2)}{2(2!)} + \frac{(x+2)^2}{4(4!)} - \frac{(x+2)^3}{8(6!)} + \frac{(x+2)^4}{16(8!)} - \frac{(x+2)^5}{32(10!)}$

b) $\sum_{n=0}^{\infty} \frac{(-1)^n (x)^n}{1.3.5 \cdots (2n+1)}$

$1 - \frac{x}{3} + \frac{x^2}{3.5} - \frac{x^3}{3.5.7} + \frac{x^4}{3.5.7.9} - \frac{x^5}{3.5.7.9.11}$

2) Find the radius of convergence and Interval of convergence of the power series.

a) $\sum_{n=0}^{\infty} (-1)^n \frac{(2x+1)^{n+1}}{3^n n!}$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{(2x+1)^{n+2}}{3^{n+1}(n+1)!} \cdot \frac{3^n n!}{(2x+1)^{n+1}} \right|$

$= \lim_{n\to\infty} \left| \frac{2x+1}{3(n+1)} \right| = 0 < 1$ so $R = \infty$

$I = (-\infty, \infty)$

b) $\sum_{n=0}^{\infty} \frac{(-1)^n (1.4.7. \cdots (3(n+1))(x-3)^n}{5^n}$

$\lim_{n\to\infty} \left| \frac{(1.4.7. \cdots (3n+1)(3n+4))(x-3)^{n+1}}{5^{n+1}} \cdot \frac{5^n}{1.4.7 \cdots (3n+1)(x-3)^n} \right|$

$= \lim_{n\to\infty} \left| \frac{(x-3)(3n+4)}{5} \right| = \infty$

so $R = 0$

$I = \{\ \}$

c) $\sum_{n=0}^{\infty} \frac{(x+1)^{2n}}{4^n(n^3+2)}$

$$\lim_{n\to\infty} \left| \frac{(x+1)^{2n+2}}{4^{n+1}((n+1)^3+2)} \cdot \frac{4^n(n^3+2)}{(x+1)^{2n}} \right|$$

$$= \lim \left| \frac{(x+1)^2(n^3+2)}{4((n+1)^3+2)} \right|$$

$$\left| \frac{(x+1)^2}{4} \right| < 1 \Rightarrow |(x+1)^2| < 4$$

$$|x+1| < 2 \qquad R = 2$$

$$I = (-3, 1)$$

test $x = -3$, $x = 1$

$$\sum_{n=0}^{\infty} \frac{(-2)^{2n}}{4^n(n^3+2)} \qquad \sum_{n=0}^{\infty} \frac{2^{2n}}{4^n(n^3+2}$$

$\sum_{n=0}^{\infty} \frac{1}{n^3+2}$ converges using the comparison test

since $\sum_{n=0}^{\infty} 1/n^3$ converges

Hence $\boxed{I = [-3, 1] \quad R = 2}$

d) $\sum_{n=1}^{\infty} \frac{(x+3)^n}{2^n (n+2)}$

$\lim_{n\to\infty} \left| \frac{(x+3)^{n+1}}{2^{n+1}(n+3)} \cdot \frac{2^n (n+2)}{(x+3)^n} \right|$

$= \lim_{n\to\infty} \left| \frac{(x+3)}{2} \frac{(n+2)}{(n+3)} \right|$

$\left|\frac{x+3}{2}\right| < 1$

$|x+3| < 2$ $\quad R = 2$

$-5 < x < -1$

$x = -5$

$\sum_{n=0}^{\infty} \frac{(-2)^n}{2^n (n+2)}$

$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n+2}$

conditionally convergent using Alternating series test

$x = -1$

$\sum_{n=0}^{\infty} \frac{2^n}{2^n (n+2)}$

$\sum_{n=0}^{\infty} \frac{1}{n+2}$

divergent by the comparison test

$$\boxed{\begin{array}{l} R = 2 \\ I = [-5, -1) \end{array}}$$

e $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}(x-1)^{2n}}{2^{n+1}(n^3+2)}$

$$\lim_{n\to\infty} \left| \frac{(-1)^{n+2}(x-1)^{2n+2}}{2^{n+2}((n+1)^3+2)} \cdot \frac{2^{n+1}(n^3+2)}{(-1)^{n+1}(x-1)^{2n}} \right|$$

$$= \lim_{n\to\infty} \left| \frac{-(x-1)^2(n^3+2)}{2((n+1)^3+2)} \right|$$

$$\left| \frac{-(x-1)^2}{2} \right| < 1$$

$$|(x-1)^2| < 2$$

$$|x-1| < \sqrt{2} \qquad \boxed{R = \sqrt{2}}$$

$$-\sqrt{2}+1 < x < 1+\sqrt{2}$$

test at $-\sqrt{2}+1$

$$\sum_{n=2}^{\infty} \frac{(-1)^{n+1}(-\sqrt{2}+1-1)^2}{2^{n+1}(n^3+2)}$$

$$= \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{2(n^3+2)}$$

Absolutely convergent by the comparison test

$x = \sqrt{2}+1$

$$\sum_{n=2}^{\infty} \frac{(-1)^{n+1}(\sqrt{2}+1-1)^2}{2^{n+1}(n^3+2)}$$

$$= \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{2(n^3+2)}$$

Absolutely convergent

Hence $R = \sqrt{2}$

$I = [-\sqrt{2}+1, 1+\sqrt{2}]$